

TED (15) – 4212

Reg.No. ....

(REVISION -2015)

Signature .....

**FOURTH SEMESTER DIPLOMA IN ENGINEERING/TECHNOLOGY**  
**INDUSTRIAL ELECTRONICS AND CONTROL DRIVES**

[Time : 3 hours ]

( Maximum marks : 100 )

**PART – A**

( Maximum marks : 10 )

Marks

I Answer the following questions in one or two sentences. Each questions carries 2 marks.

1. Define the term triggering current.
2. Write four turn on methods of SCR
3. Classify DC electric motors.
4. List four applications of servomotors
5. What is cycloconverter.

( 5x2 = 10)

**PART – B**

( Maximum marks : 30 )

II Answer any five of the following questions. Each questions carries 6 marks.

1. Explain the structure of power diode.
2. Describe the transistor analogy of SCR.
3. Draw the circuit diagram and various voltage and current waveform of single phase mid-point converter with R L load
4. Explain the principle of operation of stepper motor.
5. Describe the working principle of AC tachogenerator.
6. Enlist applications of single phase and three phase dual converter
7. Distinguish between AC and DC drives.

PART – C

( Maximum marks : 60 )

(Answer one full questions from each unit. Each full question carries 15 marks )

UNIT - I

- |     |                                                                                |   |
|-----|--------------------------------------------------------------------------------|---|
| III | (a) Explain the structure of SCR.                                              | 8 |
|     | (b) Explain the method adopted for the protection of SCR against over current. | 7 |

OR

- |    |                                                            |   |
|----|------------------------------------------------------------|---|
| IV | (a) Describe the static V-I characteristics of SCR.        | 8 |
|    | (b) Explain the structure and working principle of a DIAC. | 7 |

UNIT – II

- |   |                                                                    |   |
|---|--------------------------------------------------------------------|---|
| V | (a) Describe single phase bridge converter with relevant waveform. | 7 |
|   | (b) Explain the working of TRIAC light dimming circuit.            | 8 |

OR

- |    |                                                                                          |   |
|----|------------------------------------------------------------------------------------------|---|
| VI | (a) Explain the operation of three phase converter.                                      | 8 |
|    | (b) Draw RC halfwave trigger circuit for one SCR and discuss the function of components. | 7 |

UNIT – III

- |     |                                                                          |   |
|-----|--------------------------------------------------------------------------|---|
| VII | (a) Describe the principle of operation of DC servomotor.                | 8 |
|     | (b) Explain the principle of operation of single phase induction motors. | 7 |

OR

- |      |                                                             |   |
|------|-------------------------------------------------------------|---|
| VIII | (a) Describe the principle of operation of Universal motor. | 8 |
|      | (b) Describe the classification of DC motors.               | 7 |

UNIT –IV

- |    |                                                                             |   |
|----|-----------------------------------------------------------------------------|---|
| IX | (a) Describe the construction and working of Jones chopper circuit diagram. | 8 |
|    | (b) Explain the principle of operation of Cycloconverter.                   | 7 |

OR

- |   |                                                                                |   |
|---|--------------------------------------------------------------------------------|---|
| X | (a) Describe the speed control of induction motor by stator frequency control. | 7 |
|   | (b) Explain stepup and stepdown choppers.                                      | 8 |